

Chapter 9

BIOMOLECULES

NEET
FOCUS

The Molecules of Life

OVERVIEW

- Biomolecules are **organic molecules** produced by living organisms.
- They are the building blocks of life and perform various structural, **functional** and **regulatory** roles.
- The major classes are → **Carbohydrates**, **Proteins**, **Lipids**, **Nucleic acids**.

These molecules sustain life by providing energy, structure and information transfer.

MAJOR CLASSES OF BIOMOLECULES

Biomolecule	Elements present	Monomer / Building unit	Function	Example
Carbohydrates	C, H, O	Monosaccharides	Quick energy, structure	Glucose, Starch, Cellulose
Proteins	C, H, O, N (sometimes S)	Amino acids	Growth, repair, enzymes, hormones	Keratin, Enzymes, Antibodies
Lipids	C, H, O (P in phospholipids)	Fatty acids + Glycerol	Long-term energy storage, membranes, insulation	Fats, Oils, Phospholipids, Steroids
Nucleic acids	C, H, O, N, P	Nucleotides	Storage & transmission of genetic information	DNA, RNA

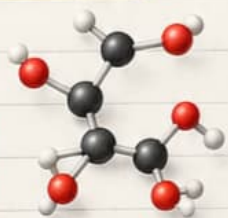
IMPORTANCE OF BIOMOLECULES

- **Carbohydrates** → Provide energy and structural support.
- **Proteins** → Catalyse reactions, build tissues, transport molecules.
- **Lipids** → Store energy, form cell membranes, act as hormones.
- **Nucleic acids** → Store and transmit genetic information.

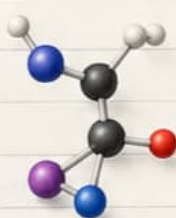


ELEMENTS IN BIOMOLECULES

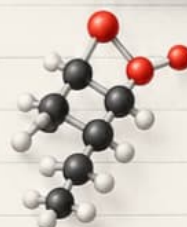
- C → Carbon
- H → Hydrogen
- O → Oxygen
- N → Nitrogen
- P → Phosphorus
- S → Sulfur



Glucose
(Carbohydrate)



Amino acid
(Protein)



Fat (Lipid)



DNA
(Nucleic acid)



Biomolecules **work together** in harmony to make life possible.

Remember:

No life without biomolecules!



1. CARBOHYDRATES

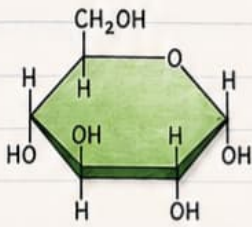
General formula
(CH₂O)_n

- Organic compounds made of C, H and O (O : H = 1 : 2).
- They are the **chief source of energy** for living organisms.
- Classified on the basis of **number of sugar units**.

CLASSIFICATION

(i) Monosaccharides

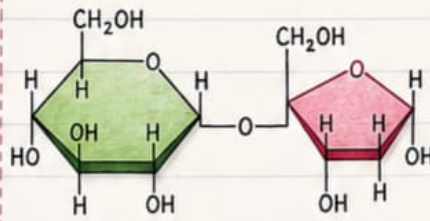
- Simplest carbohydrates
- Cannot be hydrolysed to smaller units.
- Examples: Glucose, Fructose, Ribose.



Glucose (Monosaccharide)

(ii) Oligosaccharides

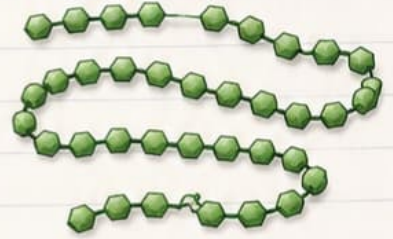
- On hydrolysis yield 2-10 monosaccharide units.
- Examples: Sucrose (2 units), Raffinose (3 units).



Sucrose (Disaccharide)
(Glucose + Fructose)

(iii) Polysaccharides

- Long chain of many (>10) monosaccharide units.
- Examples: Starch, Glycogen, Cellulose.



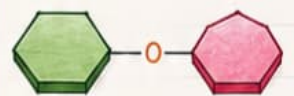
Starch (Polysaccharide)
(Helical structure)

IMPORTANT MONOSACCHARIDES

Name	Structure (Type)	Occurrence	Importance / Function
Glucose	Aldohexose (6 carbon sugar)	Fruits, honey, blood, all green plants.	- Immediate source of energy. - Stored as starch in plants and glycogen in animals.
Fructose	Ketohexose (6 carbon sugar)	Fruits, honey, sugarcane.	- Very sweet. - Part of sucrose.
Ribose	Aldopentose (5 carbon sugar)	RNA, ATP, NADP ⁺ etc.	Component of RNA, ATP, coenzymes.
Deoxyribose	Aldopentose (5 carbon sugar)	DNA.	Component of DNA.

GLYCOSIDIC BOND

The bond formed between two monosaccharides through -OH group of one sugar and -H of another.

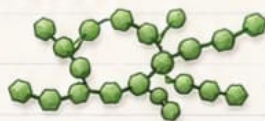


Monosaccharide - O - Monosaccharide

Glycosidic bond

POLYSACCHARIDES - EXAMPLES

- STARCH (Plants)**
Made of α-glucose.
Stored in amyloplast of plants.
- GLYCOGEN (Animals)**
Made of α-glucose.
Stored in liver and muscles.
- CELLULOSE (Plants)**
Made of β-glucose.
Structural component of plant cell wall.



Branched
(Amylopectin)



Highly branched



Unbranched
(Straight chain)

FUNCTIONS OF CARBOHYDRATES

- Provide energy (4 kcal/g).
- Stored as starch (plants) and glycogen (animals).
- Structural role - Cellulose in cell wall.
- Form parts of nucleic acids (Ribose, Deoxyribose).

Remember!

Carbohydrates are the **quick energy** givers and **structure builders** of life.



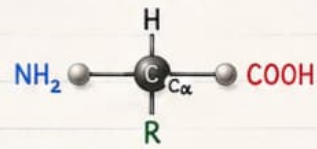
"Every molecule has a mission, Carbohydrates energize life!"

2. PROTEINS

General formula:
(C, H, O, N)
(Sometimes S)

- Proteins are **polymers** of amino acids.
- They are linked by **peptide bond**.
- 20** different amino acids occur in proteins.
- They perform **structural**, **catalytic**, **transport**, **hormonal** and **defensive** functions.

AMINO ACID - GENERAL STRUCTURE

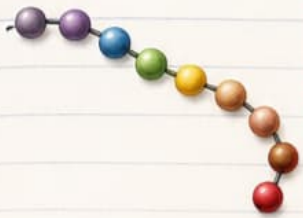


R = Side chain (varies)

LEVELS OF PROTEIN STRUCTURE

1° Primary Structure

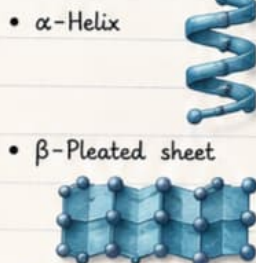
Linear sequence of amino acids.



Example: Insulin

2° Secondary Structure

Local folding of polypeptide chain.



Example: Keratin

3° Tertiary Structure

Overall 3D shape of a polypeptide.



Example: Myoglobin

4° Quaternary Structure

Association of two or more polypeptide chains.



Example: Hemoglobin (4 subunits)

FUNCTIONS OF PROTEINS

- Structural** - Keratin (hair), Collagen (connective tissue)
- Catalytic** - Enzymes speed up biochemical reactions.
- Transport** - Hemoglobin transports O_2 in blood.
- Hormonal** - Insulin regulates blood sugar level.
- Defensive** - Antibodies protect against diseases.
- Contractile** - Actin & Myosin help in muscle movement.

EXAMPLES OF PROTEINS

Protein	Source / Location	Function
Insulin	Pancreas	Regulates blood glucose level
Hemoglobin	RBCs	Transport of O_2
Antibodies	Plasma / Immune cells	Defence against pathogens
Enzymes (Amylase)	Saliva	Digestion of starch
Keratin	Hair, Nails, Skin	Structural protein
Collagen	Connective tissues	Provides strength and elasticity

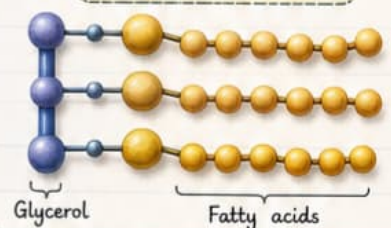
3. LIPIDS

- Lipids are heterogeneous group of **hydrophobic** biomolecules.
- Mainly composed of C, H, O and P.
- Insoluble in water but soluble in organic solvents.

TYPES OF LIPIDS

- (i) Fats & Oils (Triglycerides)
- (ii) Phospholipids
- (iii) Waxes
- (iv) Steroids

TRIGLYCERIDE (FAT)



(i) FATS & OILS

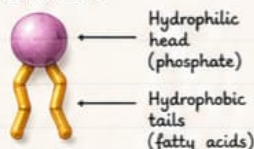
- Esters of glycerol and fatty acids.
- Fats - solid (animal origin)
- Oils - liquid (plant origin)

Function: Store energy, insulation, protection.



(ii) PHOSPHOLIPIDS

- Have glycerol, 2 fatty acids and a phosphate group.
- Amphipathic (have both hydrophilic & hydrophobic parts).
- Major component of cell membrane.



(iii) WAXES

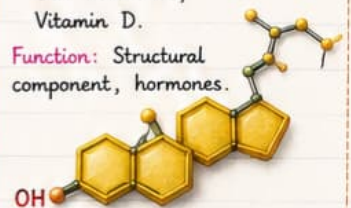
- Esters of long chain fatty acids and long chain alcohols.
- Waterproofing agents.
- Example: Beeswax, Carnauba wax.



(iv) STEROIDS

- Based on steroid nucleus (four fused carbon rings).
- Example: Cholesterol, Sex hormones, Vitamin D.

Function: Structural component, hormones.



Proteins build you, Lipids fuel you,
Carbohydrates power you!

Note: Each biomolecule has a unique role, but all are essential for life.

Nucleic acids store and transmit genetic information.

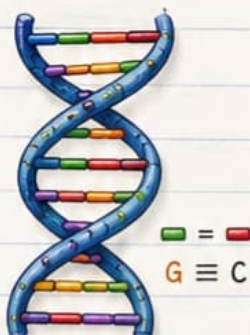
4. NUCLEIC ACIDS

Nucleotide =
Nitrogenous base
+ Pentose sugar
+ Phosphate group

- Nucleic acids are polymers of nucleotides.
- Two types - DNA (Deoxyribonucleic acid) & RNA (Ribonucleic acid).
- Built up of three components - Nitrogenous base, Pentose sugar and Phosphate group.

DNA (Deoxyribonucleic Acid)

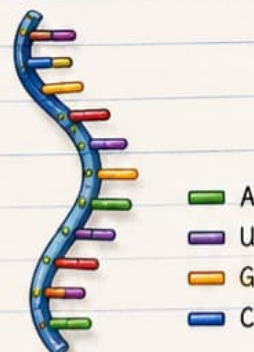
- Double stranded, right handed helical structure.
- Sugar - Deoxyribose (2' - H)
- Bases - Adenine (A), Thymine (T), Guanine (G), Cytosine (C)
- Present in nucleus, mitochondria and chloroplast.



DNA double helix

RNA (Ribonucleic Acid)

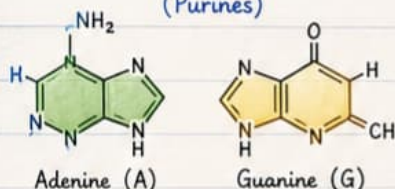
- Usually single stranded.
- Sugar - Ribose (2' - OH)
- Bases - Adenine (A), Uracil (U), Guanine (G), Cytosine (C)
- Present in nucleus and cytoplasm (ribosomes).



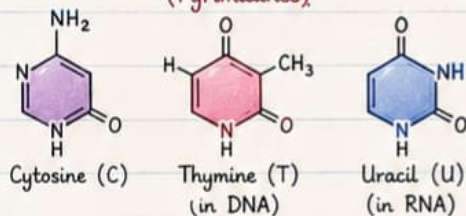
RNA single strand

COMPONENTS OF NUCLEIC ACIDS

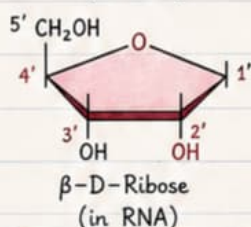
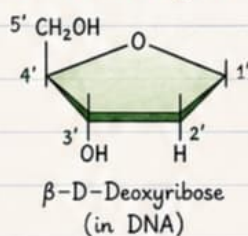
1. Nitrogenous bases (Purines)



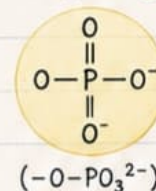
(Pyrimidines)



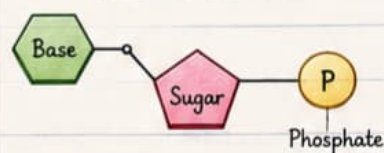
2. Pentose sugars



3. Phosphate group



NUCLEOTIDE



DIFFERENCES BETWEEN DNA AND RNA

Feature	DNA	RNA
Sugar	Deoxyribose (2' - H)	Ribose (2' - OH)
Bases	A, T, G, C	A, U, G, C (Uracil replaces Thymine)
Strands	Double stranded	Single stranded
Location	Nucleus, Mitochondria, Chloroplast	Nucleus (nucleolus), Cytoplasm (ribosomes)
Function	Storage of genetic information	Expression of genetic information
Stability	More stable	Less stable

TYPES OF RNA

- mRNA (Messenger RNA):** Carries genetic message from DNA to ribosome.
- tRNA (Transfer RNA):** Brings amino acids to ribosome during protein synthesis.
- rRNA (Ribosomal RNA):** Major structural and functional component of ribosomes.

FUNCTIONS OF NUCLEIC ACIDS

- Store genetic information and transmit it from one generation to another.
- Direct the synthesis of proteins by specifying the sequence of amino acids.
- Play essential role in heredity and variation.
- Nucleotides like ATP, NAD⁺, FAD act as energy carriers and coenzymes in metabolic reactions.

Remember!

DNA is the master information bank.
RNA is the working messenger!

From simple molecules to complex life - Biomolecules are the silent heroes that build, run and repair the machine called LIFE.

5. ENZYMES - BIOLOGICAL CATALYSTS

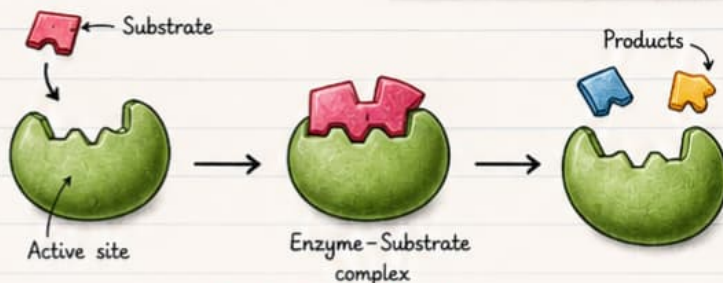
- Enzymes are **proteins** that act as **biological catalysts**.
- They increase the **speed** of biochemical reactions.
- They are highly **specific** and work under **mild conditions**.
- Most enzymes end with suffix '**-ase**' (e.g., Amylase, Lipase).
- Enzymes are not consumed in reaction and can be used repeatedly.

Why enzymes are important?

- ✓ Speed up life processes
- ✓ Save energy
- ✓ Highly specific
- ✓ Work at mild temperature and pH

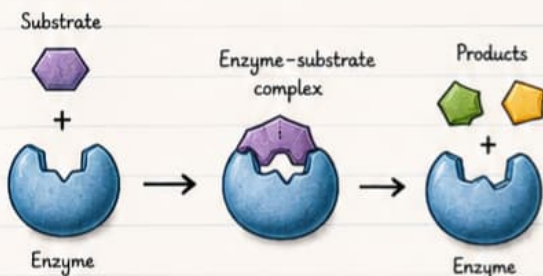
1. STRUCTURE OF ENZYME

- Enzymes have a unique 3D structure.
- Active site** - a specific region where substrate binds.
- The shape of active site is complementary to substrate (**Lock and Key model**).



Lock and Key Model

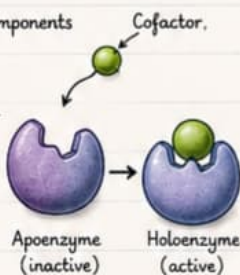
2. ENZYME ACTION (MECHANISM)



4. ENZYME COFACTORS

- Some enzymes need non-protein components called cofactors for their activity.
- Types:

- Metal ions** - e.g., Zn^{2+} , Fe^{2+} , Mg^{2+}
- Coenzymes** - organic molecules (e.g., NAD^+ , FAD, Coenzyme A)
- Prosthetic groups** - tightly bound cofactors.

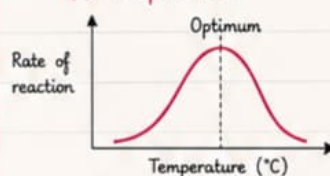


5. EXAMPLES OF ENZYMES

Enzyme	Source	Substrate	Products	Function
Amylase	Saliva	Starch	Maltose	Digests starch
Pepsin	Stomach	Proteins	Peptides	Digests proteins
Lipase	Pancreas	Fats	Fatty acids + Glycerol	Digests fats
Catalase	Liver	Hydrogen peroxide	Water + Oxygen	Detoxifies H_2O_2
DNA polymerase	All living cells	dNTPs	DNA	DNA synthesis

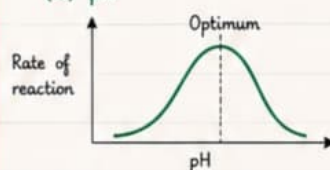
3. FACTORS AFFECTING ENZYME ACTIVITY

(i) Temperature



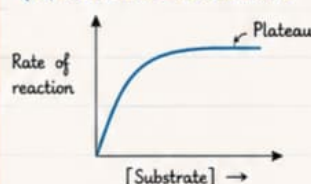
- Activity increases with temperature up to optimum.
- Above optimum, enzyme denatures.

(ii) pH



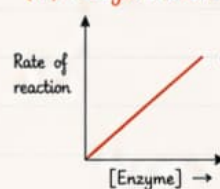
- Each enzyme has an optimum pH.
- Extreme pH changes denature the enzyme.

(iii) Substrate concentration



- Rate increases with substrate concentration till active sites are saturated.

(iv) Enzyme concentration



- Rate is directly proportional to enzyme concentration.

6. IMPORTANCE OF ENZYMES

- ★ Help in digestion of food.
- ★ Involved in energy production.
- ★ Help in blood clotting, muscle contraction, nerve impulse conduction.
- ★ Used in industries - detergents, cheese making, baking, etc.
- ★ Essential for all metabolic reactions.



Remember!

Enzymes are nature's tiny workers - small in size but do a **BIG JOB**!

Without enzymes life would be **IMPOSSIBLE**.

Enzymes make life run **SMOOTHLY**.

6. VITAMINS - ESSENTIAL ORGANIC COMPOUNDS

- Vitamins are **organic compounds** required in **small amounts** for normal growth, metabolism and maintenance of health.
- Most vitamins **cannot be synthesized** by our body, so they must be obtained from diet.
- Deficiency of vitamins causes specific **deficiency** diseases.

Key point

Vitamins do not provide energy but they act as catalyzers or co-enzymes in metabolic reactions.

VITAMINS ARE OF TWO TYPES

- Fat soluble vitamins
- A, D, E, K
- Water soluble vitamins
- B-Complex, C

FAT SOLUBLE VITAMINS

Vitamin	Sources	Functions	Deficiency disease	3D Structure / Representation
A	Carrot, Mango, Milk, Egg yolk, Fish liver oil, Green leafy vegetables.	- Essential for vision (rhodopsin). - Maintains healthy skin and mucous membranes. - Helps in growth and immunity.	Night blindness, Dry skin, Corneal ulceration, Xerophthalmia.	Retinol (Vitamin A)
D	Sunlight (UV rays), Fish liver oil, Egg yolk, Milk.	- Helps in absorption of Ca^{2+} and P. - Essential for formation of bones and teeth.	Rickets in children, Osteomalacia in adults.	Cholecalciferol (Vitamin D₃)
E	Nuts, Seeds, Vegetable oils, Wheat germ, Green leafy vegetables.	- Acts as an antioxidant. - Protects cell membranes from damage. - Essential for normal reproduction.	Sterility, Muscle weakness, Hemolysis of RBCs.	Tocopherol (Vitamin E)
K	Green leafy vegetables, Soybean, Synthesis by gut bacteria.	- Essential for blood clotting. - Helps in synthesis of prothrombin.	Haemorrhage (excessive bleeding).	Phylloquinone (Vitamin K₁)

WATER SOLUBLE VITAMINS

Vitamin	Sources	Functions	Deficiency disease	Key Representation
B-Complex (B ₁ , B ₂ , B ₃ , B ₅ , B ₆ , B ₇ , B ₉ , B ₁₂)	Whole grains, Pulses, Milk, Meat, Eggs, Yeast, Green vegetables.	- Act as coenzymes in metabolism of carbohydrates, proteins and fats. - Essential for growth, appetite and healthy nervous system.	B₁ - Beriberi B₂ - Cheilosis B₃ - Pellagra B₆ - Dermatitis B₁₂ - Pernicious anaemia B₇ - Hair loss B₉ - Megaloblastic anaemia	
C	Citrus fruits, Amla, Guava, Tomato, Potato, Green leafy vegetables.	- Important for formation of collagen. - Helps in healing of wounds. - Increases resistance against infections.	Scurvy - bleeding gums, loose teeth, poor wound healing, weak immunity.	Ascorbic acid (Vitamin C)

VITAMINS - FACTS

- Fat soluble vitamins are **stored** in the body (liver, fat tissues). Excess may cause **toxicity**.
- Water soluble vitamins are **not stored** (except B₁₂ to some extent). Excess is excreted in urine.
- Balanced diet provides all the vitamins in the right amount.

Eat a Rainbow!



Eat variety, stay healthy.

SUMMARY

- ✓ Biomolecules are the building blocks of life.
- ✓ Carbohydrates - energy & structure
- ✓ Proteins - growth & repair
- ✓ Lipids - energy storage & insulation
- ✓ Nucleic acids - store & transmit genetic information
- ✓ Biomolecules work together to sustain life.



Biomolecules may be **small**, but their impact is **HUGE!**

